GeoODE-KD: Distilling Sentence Embeddings as Teacher-Guided Geometric Dynamics

Abstract

Knowledge distillation provides an effective route for transferring the semantic capabilities of large text embedding models into compact encoders. Existing embedding distillation methods predominantly formulate knowledge transfer as *state matching*: the student is encouraged to reproduce teacher embeddings, intermediate representations, or pairwise relations at one or more discrete network depths. Such objectives specify where student representations should be located, but provide little structure on *how* they should evolve through depth. This limitation becomes particularly relevant under large teacher-student capacity gaps, where forcing shallow student layers toward highly abstract teacher representations can over-constrain their native feature organization.

We propose **GeoODE-KD**, a continuous-depth formulation of sentence embedding distillation that transfers *representation dynamics* rather than only representation states. We view normalized sentence embeddings as points on a hypersphere and construct a teacher-conditioned semantic energy that combines instance-level alignment with relational embedding geometry. The negative Riemannian gradient of this energy, run under a finite-horizon time reparameterization, defines a vector field that specifies a principled direction of semantic evolution and reaches the teacher space exactly at the final depth. Instead of introducing a Neural ODE module or an external numerical solver, we interpret the existing Transformer layers of the student as discrete integration steps of this continuous flow and regularize consecutive layers to follow the corresponding teacher-guided dynamics.

The resulting framework requires only cached final teacher embeddings, does not access teacher hidden states during student training, and introduces no additional module at inference time. We further establish that the continuous dynamics preserve embedding norms and monotonically decrease the teacher-conditioned distillation energy; in the instance-only case, the geodesic distance to the teacher contracts in closed form along continuous depth and vanishes exactly at the output layer. We will evaluate whether this dynamics-based formulation improves teacher-student transfer, representation geometry, and downstream generalization over endpoint, multi-layer, relational, and flow-based distillation baselines.

1 Introduction

Dense text embeddings have become a fundamental interface between language models and downstream applications, supporting semantic search, retrieval-augmented generation, clustering, classification, and semantic textual similarity. Recent embedding models obtain strong general-purpose representations by increasing model size, training data, and representation dimensionality. However, these improvements come with substantial inference and deployment costs, motivating the compression of large embedding models into smaller encoders through knowledge distillation (Hinton et al., 2015; Reimers and Gurevych, 2020; Xu et al., 2023; Zhang et al., 2024; Truong et al., 2025).

A common principle underlying embedding distillation is *representation matching*. Given a teacher embedding $\boldsymbol{\tau}$ and a student representation $\mathbf{z}$, the student is optimized to minimize a discrepancy such as cosine distance or mean squared error. More structured approaches enrich this supervision using contrastive signals, intermediate hidden representations, attention relations, optimal transport, or pairwise geometry (Xu et al., 2023; Truong et al., 2025). Recent work further explores depth-wise supervision. For example, ESE encourages salient semantic information to emerge in shallower representations to support depth-scalable sentence embeddings (Li et al., 2025), while TALAS anchors selected upper student layers to the teacher output and propagates relational structure toward lower layers through adjacent-layer self-distillation (Anonymous).

Despite their differences, these approaches predominantly supervise a collection of *states*:

$$\mathbf{z}^{(1)},\, \mathbf{z}^{(2)},\, \ldots,\, \mathbf{z}^{(L)}. \tag{1}$$

The corresponding objectives describe what one or more intermediate representations should resemble, but do not explicitly model the *transition* from one level of abstraction to the next. We argue that this distinction matters for embedding distillation. A shallow Transformer layer primarily captures local and lower-level linguistic structure, whereas deeper layers gradually accumulate task-relevant semantic information. Directly forcing many intermediate layers toward a fixed high-level teacher representation therefore risks destroying the student's natural hierarchical organization. Indeed, recent layer-wise embedding distillation results indicate that stronger supervision at an increasing number of student layers does not necessarily produce better transfer, particularly under substantial teacher-student capacity mismatch (Anonymous).

This observation motivates a different question:

> *Instead of specifying where every student layer should be, can the teacher specify how the student's representation should move through depth?*

The continuous-depth perspective of residual networks provides a natural language for this question. Residual transformations can be interpreted as discrete numerical approximations to an ordinary differential equation (ODE), and this connection has motivated continuous-depth neural networks and ODE-inspired Transformer architectures (Chen et al., 2018; Li et al., 2022). Under this view, network depth plays the role of time and consecutive hidden states form a discrete trajectory,

$$\mathbf{z}^{(1)} \to \mathbf{z}^{(2)} \to \cdots \to \mathbf{z}^{(L)}. \tag{2}$$

However, simply replacing Transformer layers with a generic Neural ODE does not by itself solve the embedding distillation problem. Moreover, continuous transformations have already been explored in knowledge transfer through diffusion and flow-matching mechanisms (Huang et al., 2023; Shao et al., 2024). Our goal is therefore different: we do not learn an auxiliary continuous network that transports a student output into a teacher output. Instead, we use the teacher to define a *semantic vector field*, and train the student's existing discrete layers to realize that field.

We introduce **GeoODE-KD**, a teacher-guided geometric dynamics framework for sentence embedding distillation. Our starting point is that sentence embeddings are typically compared by cosine similarity and are commonly ℓ_2-normalized. We therefore model student and teacher representations on the unit hypersphere $\mathbb{S}^{d-1}$ rather than in unconstrained Euclidean space. Given cached teacher embeddings, we construct a semantic potential energy consisting of two complementary terms. The first attracts each student sample toward its corresponding teacher representation, while the second matches the pairwise geometry of the student batch to that of the teacher. The negative Riemannian gradient of this energy defines a continuous vector field,

$$\frac{d\mathbf{Z}(t)}{dt} = -\frac{s(t)}{R(t)}\,\mathrm{grad}_{\mathbb{S}}\,\mathcal{E}\big(\mathbf{Z}(t), \mathbf{T}\big), \qquad R(t) = \int_t^1 s(u)\,du, \tag{3}$$

which describes how sentence representations should evolve toward the teacher's semantic space. The factor $1/R(t)$ is a time reparameterization that turns the gradient flow, which would only reach its minimizer as $t \to \infty$, into a finite-horizon flow that arrives at the teacher exactly at $t = 1$: in the instance-only case the trajectory is the geodesic from the student to the teacher, traversed at the pace set by $s(t)$.

Crucially, GeoODE-KD does *not* numerically integrate this equation using an external ODE solver. We assign each Transformer layer a normalized depth coordinate t_ℓ and treat consecutive student representations as discrete observations of the continuous trajectory. At each depth, one exact step of the teacher-conditioned flow, taken with the exponential map of the hypersphere, provides the desired next representation. An ODE consistency objective then aligns the actual next Transformer layer with this predicted state. Consequently, the student itself becomes the numerical realization of the teacher-guided semantic flow.

This construction has several desirable properties. First, the teacher is used only to pre-compute final sentence embeddings; teacher inference and teacher hidden states are unnecessary during student optimization. Second, teacher supervision specifies a direction of semantic improvement rather than forcing every intermediate layer to imitate the same terminal representation. Third, because the dynamics live in the tangent space of the hypersphere, they respect the geometry induced by normalized sentence embeddings. Finally, all geometric and ODE-specific machinery is used only as a training objective; deployment requires exactly the original student encoder.

Our contributions are summarized as follows:

- We formulate sentence embedding distillation as *continuous representation dynamics*, shifting the target of knowledge transfer from isolated intermediate states to the evolution of representations through network depth.

- We introduce a teacher-conditioned Riemannian flow on the embedding hypersphere. Its energy jointly captures point-wise semantic alignment and teacher relational geometry, and a finite-horizon time reparameterization makes the flow reach the teacher exactly at unit depth, yielding an explicit and interpretable semantic vector field without a learnable Neural ODE module.

- We propose an ODE consistency objective that treats the Transformer's existing layers as discrete integration steps of the teacher-guided flow. The method requires only cached teacher output embeddings and incurs no additional inference-time architecture or ODE solver.

- We provide basic theoretical guarantees for the continuous dynamics, including hyperspherical norm preservation and monotonic distillation-energy descent. In the point-wise special case, we further derive the depth profile in closed form: the geodesic distance to the teacher contracts as $R(t)/R(0)$ and vanishes at the output layer, a quantitative prediction that the experiments can test layer by layer.

2 Related Work

2.1 Knowledge Distillation for Sentence Embeddings

Knowledge distillation transfers predictive or representational knowledge from a high-capacity teacher to a compact student (Hinton et al., 2015). In sentence representation learning, Reimers and Gurevych (2020) demonstrated that sentence-level teacher representations can directly supervise a student embedding space, illustrating the effectiveness and simplicity of output-level embedding distillation. Such methods are particularly attractive when the teacher is large or accessible only through inference, since teacher embeddings can be computed offline.

Subsequent approaches enrich the distillation signal beyond direct point-wise imitation. DistillCSE incorporates knowledge distillation into contrastive sentence representation learning and studies the instability caused by high-variance teacher signals in contrastive settings (Xu et al., 2023). Jasper and Stella employ a multi-stage distillation strategy for transferring strong embedding models into more deployable representations (Zhang et al., 2024). More recently, EMO addresses cross-tokenizer embedding distillation through intra-model relational alignment and optimal transport over hidden representations (Truong et al., 2025).

Our work retains the efficiency advantage of sentence-level teacher supervision but differs in what is distilled. Rather than using the final teacher embedding solely as a static target, GeoODE-KD uses it to define a vector field that supervises how student representations evolve through depth.

2.2 Layer-wise and Geometric Representation Transfer

Intermediate supervision has long been used to expose students to richer teacher information than final-output matching alone. In embedding models, this idea is complicated by architectural and capacity differences: representations at different depths need not encode semantic information at comparable abstraction levels.

ESE addresses depth efficiency by explicitly encouraging salient information to be represented at shallower layers, enabling sentence embedding models to operate at different depths (Li et al., 2025). TALAS takes a complementary distillation perspective: selected upper student layers are aligned to a cached final teacher embedding, while lower-layer knowledge propagation is encouraged by matching pairwise geometric relations between adjacent student layers (Anonymous). Its formulation highlights an important tension: direct teacher alignment can provide high-level semantics, but excessive anchoring may over-constrain lower or intermediate student representations.

GeoODE-KD builds on the broader insight that embedding geometry contains knowledge not captured by point-wise targets. However, rather than imposing a sequence of static layer constraints or requiring each adjacent layer to preserve the same relation matrix, we define a *global potential* conditioned on the teacher geometry. Its gradient determines the desired local transition at every student depth. Thus, relational knowledge affects not only the location of embeddings but also their direction of evolution.

2.3 Continuous-Depth Networks and ODE-Based Knowledge Transfer

Neural ODEs interpret hidden representations as states of a continuous dynamical system,

$$\frac{d\mathbf{h}(t)}{dt} = f_\theta(\mathbf{h}(t), t), \tag{4}$$

whose solution replaces an explicitly specified sequence of discrete layers (Chen et al., 2018). The connection between residual networks and numerical ODE integration has also been explored for Transformers. ODE Transformer, for example, relates Transformer residual computation to numerical integration and derives architectures inspired by higher-order ODE solvers (Li et al., 2022).

Continuous transformations have separately appeared in knowledge distillation. DiffKD models the student feature as a noisy version of the teacher feature and uses diffusion-based denoising before feature matching (Huang et al., 2023). Flow Matching provides a simulation-free framework for learning continuous normalizing flows (Lipman et al., 2023), extended to Riemannian manifolds by Riemannian Flow Matching, whose reference trajectories are geodesics traversed at a prescribed schedule (Chen and Lipman, 2024), and FM-KT applies continuous flow-based transport to progressively bridge teacher-student feature or logit distributions (Shao et al., 2024). Our finite-horizon field shares the geodesic reference of Riemannian Flow Matching, but no velocity network is learned: the student's own layers are trained to realize the analytic flow.

GeoODE-KD is conceptually distinct from both continuous-depth architecture design and learned flow-based feature transport. We neither replace the student architecture with a Neural ODE block nor train an auxiliary continuous normalizing flow that maps a student output to a teacher output. The student Transformer remains unchanged. Continuous dynamics instead serve as a *training principle*: the teacher induces an analytic geometric vector field, and ordinary student layers are optimized to approximate its discretized trajectory. As a result, no ODE solver or flow module is required at inference time.

3 GeoODE-KD

3.1 Problem Formulation

Let

$$f_T : \mathcal{X} \to \mathbb{R}^{d_T}, \qquad f_S : \mathcal{X} \to \mathbb{R}^{d_S} \tag{5}$$

denote a teacher embedding model and a compact student encoder, respectively. Given an unlabeled distillation corpus

$$\mathcal{D} = \{x_i\}_{i=1}^{N}, \tag{6}$$

the objective is to transfer the semantic structure of f_T into f_S without requiring task labels.

The student contains L Transformer layers. For sentence x_i, let $\mathbf{h}_i^{(\ell)}$ denote its hidden representation at layer ℓ. A pooling operator $\mathrm{Pool}(\cdot)$ produces an intermediate sentence representation,

$$\mathbf{q}_i^{(\ell)} = \mathrm{Pool}\left(\mathbf{h}_i^{(\ell)}\right). \tag{7}$$

We focus on the common large-teacher-small-student regime where $d_T \geq d_S$. Teacher outputs are transformed once into the student dimension using a fixed linear map $P_T = P_{\mathrm{PCA}} R \in \mathbb{R}^{d_T \times d_S}$, where P_{PCA} is PCA fitted only on the cached training embeddings and $R \in O(d_S)$ is a fixed rotation inside the PCA subspace whose role is explained below:

$$\boldsymbol{\tau}_i = \mathrm{norm}\left(f_T(x_i) P_{\mathrm{PCA}} R\right), \tag{8}$$

where

$$\mathrm{norm}(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|_2 + \epsilon}. \tag{9}$$

When teacher and student dimensions are equal, P_T is the identity. PCA is not merely a convenient choice here: by the Eckart–Young theorem it is, among all rank-d_S linear maps, the one that retains the largest share of the embedding energy and therefore best preserves the Gram matrix $\mathbf{T}\mathbf{T}^\top$ on which the relational energy of Section 3.3 is defined. In our main setting ($d_T = 2560 \to d_S = 768$) the fitted map retains 93.0% of the cached embedding energy.

Gauge freedom of the endpoint. Every quantity by which a sentence encoder is evaluated, cosine similarity, Spearman correlation, linear probes, is invariant under an orthogonal transformation $\mathbf{z} \mapsto \mathbf{z}R$ of the student space, and so is the relational energy of Section 3.3. The coordinates that PCA returns are therefore an arbitrary *gauge*: any rotation of them inside the retained subspace is an equally valid target, but a point-wise objective such as Eq. (18) pins one of them. Left at the PCA gauge, the endpoint condition asks the student to rotate its entire pretrained representation space onto the teacher's principal axes, a demand unrelated to semantics: in our setting the untrained student is nearly orthogonal to such targets (mean cosine ≈ 0). We remove the gauge by fixing R once, before training, as the orthogonal Procrustes alignment (Schönemann, 1966) of the PCA coordinates to the untrained student,

$$R = \arg\min_{R \in O(d_S)} \left\| \mathbf{T}_{\mathrm{PCA}} R - \mathbf{Z}_0^{(L)} \right\|_F = UV^\top, \qquad U\Sigma V^\top = \mathrm{svd}\left(\mathbf{T}_{\mathrm{PCA}}^\top \mathbf{Z}_0^{(L)} \right), \tag{10}$$

where $\mathbf{T}_{\mathrm{PCA}}$ stacks the PCA coordinates of a subset of the corpus and $\mathbf{Z}_0^{(L)}$ the initial student's output embeddings of the same sentences. Since R is orthogonal, the Gram matrix $\mathbf{T}\mathbf{T}^\top$, the retained variance and the Eckart–Young optimality of P_{PCA} are untouched; only the arbitrary part of the endpoint condition is removed. Like P_{PCA}, R is a closed-form statistic of the cached targets and the initial student, computed once (one SVD of a $d_S \times d_S$ matrix), frozen, and discarded at inference. P_T never interacts with student optimization.

Student representations are similarly normalized,

$$\mathbf{z}_i^{(\ell)} = \mathrm{norm}\left(\mathbf{q}_i^{(\ell)} \right). \tag{11}$$

Therefore,

$$\mathbf{z}_i^{(\ell)}, \boldsymbol{\tau}_i \in \mathbb{S}^{d_S - 1}, \tag{12}$$

where $\mathbb{S}^{d_S-1}$ denotes the unit hypersphere.

For a mini-batch of size B, we stack row-wise representations as

$$\mathbf{Z}^{(\ell)} = \begin{bmatrix} \left(\mathbf{z}_1^{(\ell)}\right)^\top \\ \vdots \\ \left(\mathbf{z}_B^{(\ell)}\right)^\top \end{bmatrix} \in \mathbb{R}^{B \times d_S}, \qquad \mathbf{T} = \begin{bmatrix} \boldsymbol{\tau}_1^\top \\ \vdots \\ \boldsymbol{\tau}_B^\top \end{bmatrix}. \tag{13}$$

3.2 From Discrete Layer Matching to Continuous Semantic Dynamics

Conventional multi-layer distillation can be written schematically as

$$\mathcal{L}_{\text{layer}} = \sum_{\ell \in \mathcal{A}} D\left(\mathbf{Z}^{(\ell)}, \mathbf{T} \right), \tag{14}$$

where $\mathcal{A}$ denotes a selected set of student layers. Equation (14) treats each supervised layer as an independent state-matching problem.

We instead introduce a normalized depth coordinate

$$t_\ell = \frac{\ell}{L}, \qquad \ell \in \{1, \ldots, L\}, \tag{15}$$

and interpret

$$\mathbf{Z}^{(\ell)} \approx \mathbf{Z}(t_\ell) \tag{16}$$

as samples from an underlying continuous semantic trajectory.

Our objective is to construct a teacher-conditioned differential equation

$$\frac{d\mathbf{Z}(t)}{dt} = F\left(\mathbf{Z}(t), \mathbf{T}, t\right) \tag{17}$$

and train the discrete Transformer layers to approximate this flow.

3.3 Teacher-Conditioned Semantic Energy

The vector field in Eq. (17) should satisfy two properties. First, each sentence representation should progressively move toward the semantic target supplied by its teacher embedding. Second, the student should recover the relational geometry of the teacher embedding space rather than merely matching samples independently.

We encode these requirements through a teacher-conditioned energy function.

Instance-level semantic energy. For two points on the unit hypersphere, the intrinsic notion of discrepancy is the geodesic (great-circle) distance $d_g(\mathbf{z}, \boldsymbol{\tau}) = \arccos(\mathbf{z}^\top \boldsymbol{\tau})$. We define

$$\mathcal{E}_{\text{sem}}(\mathbf{Z}, \mathbf{T}) = \frac{1}{2B} \sum_{i=1}^{B} d_g^2\left(\mathbf{z}_i, \boldsymbol{\tau}_i\right). \tag{18}$$

The squared geodesic distance is the natural potential on a Riemannian manifold: its negative Riemannian gradient is the logarithmic map $\mathrm{Log}_{\mathbf{z}}(\boldsymbol{\tau})$, the tangent vector that points along the geodesic to $\boldsymbol{\tau}$ and whose length is d_g. Near the teacher it agrees with the cosine distance, $\frac{1}{2}d_g^2 = 1 - \mathbf{z}^\top \boldsymbol{\tau} + O(d_g^4)$, but its gradient does not vanish for far-away states, which is what will allow the flow below to reach the teacher in finite depth.

Relational geometric energy. Sentence embeddings are primarily useful through relative similarities. We therefore construct batch-wise Gram matrices

$$\mathbf{G}_S = \mathbf{Z}\mathbf{Z}^\top, \qquad \mathbf{G}_T = \mathbf{T}\mathbf{T}^\top. \tag{19}$$

Their (i, j)-th entries correspond to pairwise cosine similarities.

The geometric energy is

$$\mathcal{E}_{\text{geo}}(\mathbf{Z}, \mathbf{T}) = \frac{1}{B^2} \left\| \mathbf{Z}\mathbf{Z}^\top - \mathbf{T}\mathbf{T}^\top \right\|_F^2. \tag{20}$$

The total teacher-conditioned potential is

$$\boxed{\mathcal{E}(\mathbf{Z}, \mathbf{T}) = \alpha \mathcal{E}_{\text{sem}}(\mathbf{Z}, \mathbf{T}) + \beta \mathcal{E}_{\text{geo}}(\mathbf{Z}, \mathbf{T})} \tag{21}$$

with $\alpha, \beta \geq 0$.

Importantly, Eq. (21) is not directly minimized at every student layer. Instead, it defines the continuous dynamics that student layers are trained to realize.

3.4 Teacher-Guided Finite-Horizon Riemannian Flow

Because the embeddings lie on $\mathbb{S}^{d_S-1}$, unconstrained Euclidean motion is undesirable: an arbitrary update can alter vector norms and introduce radial components that are irrelevant to cosine geometry. We therefore construct a Riemannian flow on the product of hyperspheres, driven by the negative gradient of $\mathcal{E}$ and reparameterized in time so that it reaches the teacher within the unit depth interval.

For an embedding $\mathbf{z}$, the tangent space is

$$T_{\mathbf{z}}\mathbb{S}^{d_S-1} = \{\mathbf{v} \in \mathbb{R}^{d_S} : \mathbf{z}^\top \mathbf{v} = 0\}. \tag{22}$$

The orthogonal projection of a Euclidean vector $\mathbf{u}$ onto this tangent space is

$$\Pi_{\mathbf{z}}(\mathbf{u}) = \mathbf{u} - (\mathbf{z}^\top \mathbf{u})\mathbf{z}. \tag{23}$$

For the batch representation, $\Pi_{\mathbf{Z}}$ applies Eq. (23) row-wise.

The negative Riemannian gradient of the semantic energy is the row-wise logarithmic map of the hypersphere,

$$-\operatorname{grad}_{\mathbb{S}} \mathcal{E}_{\text{sem}} = \frac{1}{B}\operatorname{Log}_{\mathbf{Z}}(\mathbf{T}), \qquad \operatorname{Log}_{\mathbf{z}}(\boldsymbol{\tau}) = \frac{d_g(\mathbf{z}, \boldsymbol{\tau})}{\sin d_g(\mathbf{z}, \boldsymbol{\tau})}\Pi_{\mathbf{z}}(\boldsymbol{\tau}), \tag{24}$$

which is already tangent to the sphere, whereas the Euclidean gradient of the geometric term satisfies

$$\nabla_{\mathbf{Z}}\mathcal{E}_{\text{geo}} = \frac{4}{B^2}\left(\mathbf{Z}\mathbf{Z}^\top - \mathbf{T}\mathbf{T}^\top\right)\mathbf{Z}. \tag{25}$$

We therefore define the tangent descent direction

$$\mathbf{V}(\mathbf{Z}, \mathbf{T}) = \alpha \operatorname{Log}_{\mathbf{Z}}(\mathbf{T}) - \frac{4\beta}{B}\Pi_{\mathbf{Z}}\left[\left(\mathbf{Z}\mathbf{Z}^\top - \mathbf{T}\mathbf{T}^\top\right)\mathbf{Z}\right] = -B \operatorname{grad}_{\mathbb{S}} \mathcal{E}(\mathbf{Z}, \mathbf{T}), \tag{26}$$

$$F(\mathbf{Z}, \mathbf{T}, t) = \frac{s(t)}{R(t)}\mathbf{V}(\mathbf{Z}, \mathbf{T}), \qquad R(t) = \int_t^1 s(u)\, du. \tag{27}$$

The factor B in Eq. (26) removes the $1/B$ that differentiating a batch mean introduces, so that the speed of the flow does not depend on the batch size; it does not change the direction.

The continuous teacher-guided dynamics are then

$$\boxed{\frac{d\mathbf{Z}(t)}{dt} = F(\mathbf{Z}(t), \mathbf{T}, t) = -\frac{s(t)}{R(t)}\, B \operatorname{grad}_{\mathbb{S}} \mathcal{E}\left(\mathbf{Z}(t), \mathbf{T}\right)} \tag{28}$$

where $s(t) \geq 0$ controls how teacher guidance is distributed over depth and $R(t)$ is the guidance that remains to be spent after depth t.

Why a finite-horizon reparameterization. A plain gradient flow $\dot{\mathbf{Z}} = -s(t) \operatorname{grad} \mathcal{E}$ converges to the teacher only as $t \to \infty$, because the gradient vanishes at the minimizer. Over the unit depth interval it can cover at most $\int_0^1 s(t)\, \|\mathbf{V}\|\, dt$ of geodesic length, whereas a student whose first layer is nearly orthogonal to the teacher has to travel about $\pi/2$. With $s(t) = t$ this is at most 0.5 against 1.6 radians, so such a flow cannot reach the teacher, and a consistency loss built on it can only be satisfied by moving slowly at every depth and leaving the output layer to make the jump alone. The reparameterization by $1/R(t)$ is the substitution $d\tau = s(t)\, dt/R(t)$, under which the gradient flow is run for infinite τ-time while t only spans $[0, 1]$. Corollary 1 below shows that the resulting instance-only trajectory is exactly the geodesic to the teacher, traversed so that the remaining distance is proportional to the remaining guidance $R(t)$.

Depth-dependent semantic guidance. We use

$$s(t) = t \tag{29}$$

as the default schedule. This choice introduces weak teacher guidance at shallow depths and progressively strengthens it toward the output layers. The design reflects the intuition that shallow representations should retain flexibility to extract lower-level linguistic information, while deeper layers should become increasingly organized by the teacher's semantic geometry.

With this choice $R(t) = (1 - t^2)/2$, so the fraction of the remaining geodesic covered between consecutive depths grows with depth (Eq. (31)). More general schedules, e.g., $s(t) = t^p$ with $R(t) = (1 - t^{p+1})/(p + 1)$, or a constant $s(t) = 1$ with $R(t) = 1 - t$, can be considered as ablations; none introduces learned parameters, and all of them reach the teacher at $t = 1$.

3.5 Student Layers as Discrete ODE Steps

A central design choice of GeoODE-KD is that Eq. (28) is *not* solved using a black-box ODE solver. Instead, the student Transformer itself provides the discrete trajectory.

For consecutive student depths,

$$\Delta t = t_{\ell+1} - t_\ell = \frac{1}{L}. \tag{30}$$

The time warp in Eq. (27) is integrated exactly over the layer interval rather than evaluated at its left endpoint. Since $\dot{R} = -s$, the warped time elapsed between two depths is $\int_{t_\ell}^{t_{\ell+1}} s(u)/R(u)\, du = \log\big(R(t_\ell)/R(t_{\ell+1})\big)$, and the corresponding step fraction is

$$\rho_\ell = 1 - \exp\left(-\int_{t_\ell}^{t_{\ell+1}} \frac{s(u)}{R(u)}\, du\right) = 1 - \frac{R(t_{\ell+1})}{R(t_\ell)} \in (0, 1], \tag{31}$$

which is the fraction of the remaining geodesic to the target that the flow covers between depths t_ℓ and $t_{\ell+1}$. For $s(t) = t$, $\rho_\ell = (2\ell + 1)/(L^2 - \ell^2)$, and $\rho_{L-1} = 1$ for every schedule.

Given $\mathbf{Z}^{(\ell)}$, one step of the teacher-guided dynamics is

$$\widetilde{\mathbf{Z}}^{(\ell+1)} = \mathrm{Exp}_{\mathbf{Z}^{(\ell)}}\left(\rho_\ell\, \mathbf{V}(\mathbf{Z}^{(\ell)}, \mathbf{T})\right), \tag{32}$$

where Exp is the exponential map of the hypersphere, applied row-wise and available in closed form:

$$\mathrm{Exp}_{\mathbf{z}}(\mathbf{v}) = \cos\left(\|\mathbf{v}\|\right) \mathbf{z} + \sin\left(\|\mathbf{v}\|\right) \frac{\mathbf{v}}{\|\mathbf{v}\|}. \tag{33}$$

Unlike a first-order retraction such as row-wise normalization of $\mathbf{Z} + \mathbf{v}$, which rotates by $\arctan \|\mathbf{v}\|$ instead of $\|\mathbf{v}\|$ and would fall short on the large final step, the exponential map is exact for every step size. In the instance-only case ($\beta = 0$), Eq. (32) is precisely spherical linear interpolation, $\widetilde{\mathbf{z}}^{(\ell+1)} = \mathrm{slerp}(\mathbf{z}^{(\ell)}, \boldsymbol{\tau}; \rho_\ell)$, so the predicted state for the output layer is the teacher itself.

The actual Transformer produces $\mathbf{Z}^{(\ell+1)}$. We regularize it toward the ODE-predicted next state:

$$\mathcal{L}_{\mathrm{dyn}} = \frac{1}{L-1} \sum_{\ell=1}^{L-1} D_{\cos}\left(\mathbf{Z}^{(\ell+1)}, \mathrm{sg}\left[\widetilde{\mathbf{Z}}^{(\ell+1)}\right]\right), \tag{34}$$

where sg$[\cdot]$ denotes stop-gradient and

$$D_{\cos}(\mathbf{A}, \mathbf{B}) = \frac{1}{B} \sum_{i=1}^{B} \left(1 - \mathbf{a}_i^\top \mathbf{b}_i\right). \tag{35}$$

The stop-gradient operation makes $\widetilde{\mathbf{Z}}^{(\ell+1)}$ a local dynamics target computed from the current trajectory, preventing the model from trivially modifying both sides of the consistency objective simultaneously. We will separately ablate full-gradient dynamics.

Equation (34) creates the main distinction between GeoODE-KD and static multi-layer distillation:

$$\text{State matching:} \qquad \mathbf{Z}^{(\ell)} \to \mathbf{T}, \tag{36}$$

$$\text{GeoODE-KD:} \qquad \mathbf{Z}^{(\ell)} \to \mathbf{Z}^{(\ell+1)} \quad \text{according to } F(\mathbf{Z}^{(\ell)}, \mathbf{T}, t_\ell). \tag{37}$$

The teacher thus supervises the student's *velocity through depth* instead of demanding that every intermediate representation imitate the same terminal target: each layer is asked to cover a prescribed fraction ρ_ℓ of what remains, and the fractions compose to the whole geodesic.

3.6 Endpoint Semantic Distillation

While $\mathcal{L}_{\text{dyn}}$ determines the local trajectory, the student output should ultimately remain close to the teacher endpoint. We therefore apply direct distillation only at the final student layer:

$$\boxed{\mathcal{L}_{\text{end}} = \frac{1}{B}\sum_{i=1}^{B}\left(1 - \left(\mathbf{z}_i^{(L)}\right)^\top \boldsymbol{\tau}_i\right)}. \tag{38}$$

Unlike conventional multi-layer teacher anchoring, no intermediate student layer is directly forced to equal the final teacher embedding. Intermediate supervision occurs only through the vector field in Eq. (27). Because $\rho_{L-1} = 1$, the dynamics target for the output layer already coincides with $\boldsymbol{\tau}_i$ when $\beta = 0$; $\mathcal{L}_{\text{end}}$ is therefore the boundary condition of the same flow rather than a competing objective, and it remains the only term that anchors the output when the relational term bends the final step.

3.7 Contrastive Representation Regularization

Pure teacher imitation may transfer undesirable concentration or anisotropy from the teacher embedding distribution. We therefore retain a standard unsupervised contrastive objective at the final layer.

For each input x_i, two independent dropout masks produce normalized representations $\mathbf{z}_i^{(L,a)}$ and $\mathbf{z}_i^{(L,b)}$. We use

$$\mathcal{L}_{\text{ctr}} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp\left(\text{sim}(\mathbf{z}_i^{(L,a)}, \mathbf{z}_i^{(L,b)})/\tau_c\right)}{\sum_{j=1}^{B}\exp\left(\text{sim}(\mathbf{z}_i^{(L,a)}, \mathbf{z}_j^{(L,b)})/\tau_c\right)}, \tag{39}$$

where τ_c is the contrastive temperature.

Importantly, this objective is applied only to the final representation. We avoid imposing contrastive targets independently at every layer, which would confound the role of continuous dynamics.

3.8 Overall Training Objective

The final GeoODE-KD objective is

$$\boxed{\mathcal{L}_{\text{total}} = \lambda_{\text{end}}\mathcal{L}_{\text{end}} + \lambda_{\text{dyn}}\mathcal{L}_{\text{dyn}} + \lambda_{\text{ctr}}\mathcal{L}_{\text{ctr}}}. \tag{40}$$

The method introduces two conceptually distinct hyperparameters inside the semantic potential: α controls point-wise teacher attraction and β controls relational geometric preservation. For the initial implementation, we set $\alpha = 1$ and tune only β, while fixing $\lambda_{\text{end}} = \lambda_{\text{dyn}} = 1$ and tuning λ_{ctr} on a validation split.

We emphasize the separation between Eq. (21) and Eq. (40). The energy $\mathcal{E}$ is not simply another layer-wise training loss. Instead, it defines the vector field that specifies the desired *direction of change*, whereas $\mathcal{L}_{\text{dyn}}$ measures whether the Transformer's actual layer transition follows that direction.

3.9 Theoretical Properties

We next state basic properties that motivate the proposed dynamics.

Proposition 1: Hyperspherical norm preservation. Consider a single trajectory $\mathbf{z}(t) \in \mathbb{S}^{d-1}$ governed by

$$\dot{\mathbf{z}} = \Pi_{\mathbf{z}}(\mathbf{u}). \tag{41}$$

Then

$$\frac{d}{dt}\|\mathbf{z}(t)\|_2^2 = 0. \tag{42}$$

Proof. By construction, $\Pi_{\mathbf{z}}(\mathbf{u})$ lies in the tangent space of the hypersphere and therefore

$$\mathbf{z}^\top \Pi_{\mathbf{z}}(\mathbf{u}) = 0. \tag{43}$$

Consequently,

$$\frac{d}{dt}\|\mathbf{z}\|_2^2 = 2\mathbf{z}^\top \dot{\mathbf{z}} = 0. \qquad \square \tag{44}$$

Thus, the continuous teacher-guided dynamics alter semantic direction without introducing an irrelevant radial component.

Proposition 2: Monotonic teacher-conditioned energy descent. Let the continuous trajectory follow Eq. (28),

$$\dot{\mathbf{Z}} = -\frac{s(t)}{R(t)}\, B\, \mathrm{grad}_{\mathbb{S}}\, \mathcal{E}(\mathbf{Z}, \mathbf{T}), \tag{45}$$

where $s(t) \geq 0$ and $t < 1$. Then

$$\frac{d}{dt}\mathcal{E}(\mathbf{Z}(t), \mathbf{T}) \leq 0. \tag{46}$$

Proof. Using the Riemannian chain rule,

$$\frac{d\mathcal{E}}{dt} = \left\langle \mathrm{grad}_{\mathbb{S}}\, \mathcal{E}, \dot{\mathbf{Z}} \right\rangle \tag{47}$$

$$= -\frac{B\, s(t)}{R(t)}\, \|\mathrm{grad}_{\mathbb{S}}\, \mathcal{E}\|_F^2 \leq 0. \qquad \square \tag{48}$$

Hence, the ideal continuous trajectory monotonically reduces the combined semantic and relational discrepancy from the teacher.

Corollary 1: Geodesic contraction and exact arrival under the point-wise flow. Consider $\beta = 0$, $\alpha = 1$, and a single normalized teacher-student pair $(\boldsymbol{\tau}, \mathbf{z})$. The dynamics reduce to

$$\dot{\mathbf{z}} = \frac{s(t)}{R(t)} \operatorname{Log}_{\mathbf{z}}(\boldsymbol{\tau}). \tag{49}$$

Let $\theta(t) = d_g(\mathbf{z}(t), \boldsymbol{\tau})$. Since $\operatorname{Log}_{\mathbf{z}}(\boldsymbol{\tau})$ is the unit-speed geodesic direction scaled by θ, the trajectory stays on the geodesic from $\mathbf{z}(0)$ to $\boldsymbol{\tau}$ and

$$\dot{\theta}(t) = -\frac{s(t)}{R(t)}\,\theta(t), \qquad \dot{R}(t) = -s(t) \quad \Longrightarrow \quad \frac{d}{dt}\log\theta = \frac{d}{dt}\log R. \tag{50}$$

Hence

$$\boxed{\theta(t) = \theta(0)\,\frac{R(t)}{R(0)}} \qquad \text{and} \qquad \theta(1) = 0. \tag{51}$$

The geodesic distance to the teacher therefore decreases monotonically, in proportion to the guidance that remains, and vanishes exactly at unit depth: with the default $s(t) = t$, $\theta(t) = \theta(0)(1 - t^2)$. Cosine similarity $\cos\theta(t)$ increases monotonically as a consequence. Equation (51) is a quantitative prediction for the depth profile of a student that realizes the flow, and the discrete steps of Eq. (32) reproduce it exactly at the sampled depths: $\theta(t_{\ell+1}) = (1 - \rho_\ell)\theta(t_\ell)$. When relational geometry is included, individual distances need not follow Eq. (51), but Proposition 2 guarantees monotonic descent of the joint semantic-geometric energy.

3.10 Training Algorithm

3.11 Training and Inference Complexity

Teacher inference is entirely offline. For a corpus of N examples, only one final teacher embedding per training example is stored. During student training, GeoODE-KD does not require teacher hidden states, attention maps, or parallel teacher forward passes.

For a batch of B samples and embedding dimension d_S, computing the relational component at one layer requires $\mathcal{O}(B^2 d_S)$ operations for the Gram matrix and $\mathcal{O}(B^2)$ temporary relation memory. Across L student layers, the dynamics objective therefore adds approximately

$$\mathcal{O}(LB^2 d_S) \tag{52}$$

training computation. This cost is independent of the size of the teacher.

At inference time, all teacher-conditioned components are removed:

$$x \xrightarrow{f_S} \mathbf{z}^{(L)}. \tag{53}$$

No PCA target projection, gauge rotation, Gram matrix, semantic energy, vector field, ODE solver, or auxiliary network is evaluated. GeoODE-KD therefore has the same inference architecture and asymptotic inference cost as its student backbone.

3.12 Relationship to Alternative Distillation Objectives

GeoODE-KD can be understood by contrasting four forms of supervision.

Endpoint KD.

$$\mathbf{Z}^{(L)} \to \mathbf{T}. \tag{54}$$

Only the final state is constrained.

Algorithm 1 GeoODE-KD Training

Require: Unlabeled corpus $\mathcal{D}$, teacher f_T, student f_S with L layers, α, β, $\lambda_{\text{end}}, \lambda_{\text{dyn}}, \lambda_{\text{ctr}}$

Run f_T once over $\mathcal{D}$ and cache teacher embeddings
Fit fixed PCA map P_{PCA} on cached teacher embeddings
Fix the gauge R by Procrustes against the initial student, Eq. (10)
Transform and normalize all cached targets using Eq. (8)
for each training mini-batch $\mathcal{B}$ **do**
    Retrieve cached teacher matrix $\mathbf{T}$
    Run the student and obtain $\{\mathbf{Z}^{(\ell)}\}_{\ell=1}^{L}$
    $\mathcal{L}_{\text{dyn}} \leftarrow 0$
    for $\ell = 1, \ldots, L-1$ **do**
        $t_\ell \leftarrow \ell/L$
        Compute $\mathbf{G}_S \leftarrow \mathbf{Z}^{(\ell)}(\mathbf{Z}^{(\ell)})^\top$
        Compute $\mathbf{G}_T \leftarrow \mathbf{T}\mathbf{T}^\top$
        Compute the tangent direction $\mathbf{V}(\mathbf{Z}^{(\ell)}, \mathbf{T})$ using Eq. (26)
        $\rho_\ell \leftarrow 1 - R(t_{\ell+1})/R(t_\ell)$ using Eq. (31)
        Predict $\widetilde{\mathbf{Z}}^{(\ell+1)} \leftarrow \operatorname{Exp}_{\mathbf{Z}^{(\ell)}}(\rho_\ell \mathbf{V})$ using Eq. (32)
        Accumulate ODE consistency using Eq. (34)
    end for
    Compute endpoint loss $\mathcal{L}_{\text{end}}$ using Eq. (38)
    Obtain a second dropout view and compute $\mathcal{L}_{\text{ctr}}$
    Compute $\mathcal{L}_{\text{total}}$ using Eq. (40)
    Update student parameters with AdamW
end for

Static multi-layer KD.

$$\mathbf{Z}^{(\ell)} \to \mathbf{T} \qquad \forall \ell \in \mathcal{A}. \tag{55}$$

Multiple states are constrained toward a common endpoint.

Adjacent relational self-distillation.

$$\mathbf{G}^{(\ell)} \to \mathbf{G}^{(\ell+1)}. \tag{56}$$

Local student geometries are encouraged to remain consistent across depth.

GeoODE-KD.

$$\boxed{\mathbf{Z}^{(\ell+1)} \approx \operatorname{Exp}_{\mathbf{Z}^{(\ell)}}\left[\rho_\ell\, \mathbf{V}(\mathbf{Z}^{(\ell)}, \mathbf{T})\right]}. \tag{57}$$

Here, the teacher specifies a *directional transformation* at each depth. The student is not required to arrive at the final teacher representation prematurely; it is required to cover, at each depth, the prescribed fraction of the remaining geodesic to the teacher, along a direction that continuously decreases the teacher-conditioned potential.

Research hypothesis. Our central hypothesis is that a compact student benefits more from learning a *semantic trajectory* than from repeatedly imitating a high-capacity teacher state. If correct, GeoODE-KD should exhibit three observable behaviors. First, student-teacher semantic discrepancy should decrease more smoothly through network depth than under static multi-layer distillation; Corollary 1 makes this quantitative, predicting a geodesic distance profile proportional to $R(t_\ell)$ rather than one that stays flat and collapses in the last layers. Second, the teacher-student relational geometry gap should contract progressively across layers rather than only at the output. Third, these improvements should translate into better downstream embedding quality without requiring stronger inference-time computation. The experiments will directly test each of these hypotheses rather than evaluating downstream accuracy alone.

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